1. LearnMate: Turn Study Materials into Personalised Learning

Problem statement:
Students often have access to textbooks, notes, and presentations but struggle to identify what they do not understand and what to study next. Build an AI-powered learning companion that turns uploaded course material into a personalised learning and revision experience.

Minimum requirements:

- Accept a PDF or pasted study material.
- Explain concepts in simple language, with references to the relevant pages or passages.
- Generate a short diagnostic quiz and identify weak topics from the answers.
- Create a personalised revision plan and follow-up practice questions.
- Save quiz results and show progress across attempts.
- Clearly indicate when an answer is unavailable in the uploaded material.

Google AI Studio application:
Use Gemini to interpret learning material, generate explanations and quizzes, and adapt practice to the student's performance.

24-hour scope:
Support one subject, documents up to 20 pages, and one complete learning journey.

Expected demonstration:
Upload a chapter, complete a quiz, receive targeted practice, and show how the revision plan changes.

Optional extension:
Add Hindi explanations or spoken revision summaries.

2. WasteWise: Help People Sort Everyday Waste Correctly

Problem statement:
People may be unsure how to dispose of everyday items, particularly when packaging contains mixed materials or local collection rules differ. Build an AI-powered assistant that identifies an item and explains how to dispose of it using a supplied local waste-management guide.

Minimum requirements:

- Accept an item photograph or a text description.
- Suggest the item's material and ask clarifying questions when necessary.
- Recommend the appropriate waste category using the supplied guide.
- Explain any preparation steps supported by that guide.
- Display the rule or passage supporting the recommendation.
- Let users correct the identification and save a disposal history.

Google AI Studio application:
Use Gemini's image understanding to interpret photographs and its document understanding to explain the relevant disposal rules.

24-hour scope:
Support one campus or locality, one disposal guide, and 15–20 common household items.

Expected demonstration:
Process several items, including mixed packaging and an unclear photograph, showing how the app handles uncertainty.

Optional extension:
Add a household waste dashboard or a recycling knowledge challenge.

3. CampusConnect: Turn Student Complaints into Trackable Action

Problem statement:
Campus maintenance complaints often arrive through scattered messages with incomplete information. Students struggle to track progress, while staff must manually identify the issue and responsible department. Build an AI-powered reporting system that converts informal complaints into clear, actionable tickets.

Minimum requirements:

- Accept a written complaint, location, and optional photograph.
- Extract the issue, summarise it, and request missing information.
- Suggest a category and responsible department using a supplied routing guide.
- Flag potentially duplicate reports for staff review.
- Provide a staff dashboard to assign tickets and update their status.
- Show students a ticket number and progress history.

Google AI Studio application:
Use Gemini to interpret complaints, analyse supporting photographs, and generate structured ticket details.

24-hour scope:
Support one campus and four categories: plumbing, electricity, cleanliness, and internet connectivity.

Expected demonstration:
Submit an incomplete complaint, supply the missing details, review a possible duplicate, and track the ticket through resolution.

Optional extension:
Add Hindi complaint entry or recurring-issue analysis.

4. ShopLedger: Convert Receipts into Useful Business Records

Problem statement:
Small business owners often keep purchase records as paper receipts or photographs,

making it difficult to organise expenses and understand spending. Build an AI-powered assistant that converts receipts into editable records and produces clear, evidence-based spending summaries.

Minimum requirements:

- Upload receipt photographs or PDFs.
- Extract the vendor, date, items, and total into an editable form.
- Flag unclear or missing fields for user review.
- Categorise confirmed expenses and identify possible duplicate receipts.
- Calculate spending totals and category breakdowns using application code.
- Answer questions about confirmed records and export them as CSV.

Google AI Studio application:
Use Gemini to extract receipt information, suggest categories, and explain calculated spending patterns in plain language.

24-hour scope:
Support one business, one currency, and approximately 20 synthetic receipts.

Expected demonstration:
Upload receipts, correct an extraction error, review a duplicate, and generate a spending summary linked to the underlying records.

Optional extension:
Add budget comparisons or support another receipt language.